

# Customer Segmentation Project

Segments customers by demographics and purchase behavior using K-means clustering (Python) with a full implementation guide for reproducing and visualizing the results in Power BI.

## Files in this package

| File                                                   | Purpose                                                                                                                                   |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <code>Customer_Segmentation_PowerBI_Report.docx</code> | Main deliverable<br>— objective,<br>methodology,<br>DAX/Power Query<br>steps, segment<br>results,<br>dashboard layout,<br>recommendations |
| <code>customer_data.csv</code>                         | Raw synthetic<br>dataset (600<br>customers).<br>Import this into<br>Power BI to<br>reproduce the<br>pipeline from<br>scratch              |
| <code>customer_data_segmented.csv</code>               | Same data with<br><code>Segment</code> /<br><code>SegmentName</code><br>columns already<br>added — use this                               |

| File                             | Purpose                                                                              |
|----------------------------------|--------------------------------------------------------------------------------------|
|                                  | for fast visual building                                                             |
| <code>segment_summary.csv</code> | Aggregated averages per segment (size, income, RFM metrics)                          |
| <code>cluster_analysis.py</code> | Python/scikit-learn script that generated the clusters and every chart in the report |

## How to reproduce

### Python:

```
pip install pandas numpy scikit-learn matplotlib
seaborn
python cluster_analysis.py
```

Outputs `customer_data_segmented.csv`, `segment_summary.csv`, and the chart PNGs.

### Power BI:

1. Get Data → Text/CSV → `customer_data_segmented.csv` (fastest) or `customer_data.csv` (to build clustering yourself via Power BI's "Find clusters" or a Python visual – steps in Section 3 of the report).
2. Add the RFM DAX measures from Section 3.2 of the report.
3. Build visuals per the dashboard layout in Section 6.

## Segments identified

| Segment          | Size | Profile                                           |
|------------------|------|---------------------------------------------------|
| Champions        | 12%  | Highest spend, very recent, low discount reliance |
| Loyal Regulars   | 29%  | Consistent moderate spend, reasonably recent      |
| At Risk          | 38%  | Long time since last purchase, discount-sensitive |
| New / Occasional | 21%  | Low frequency, high online share                  |

## Notes

- Dataset is synthetic, built to reflect realistic retail purchase patterns — swap in a real dataset (same column names) to rerun everything as-is.
- k=5 was tested but converged to 4 well-separated clusters (validated via elbow method + silhouette score, see report Section 3.4).